

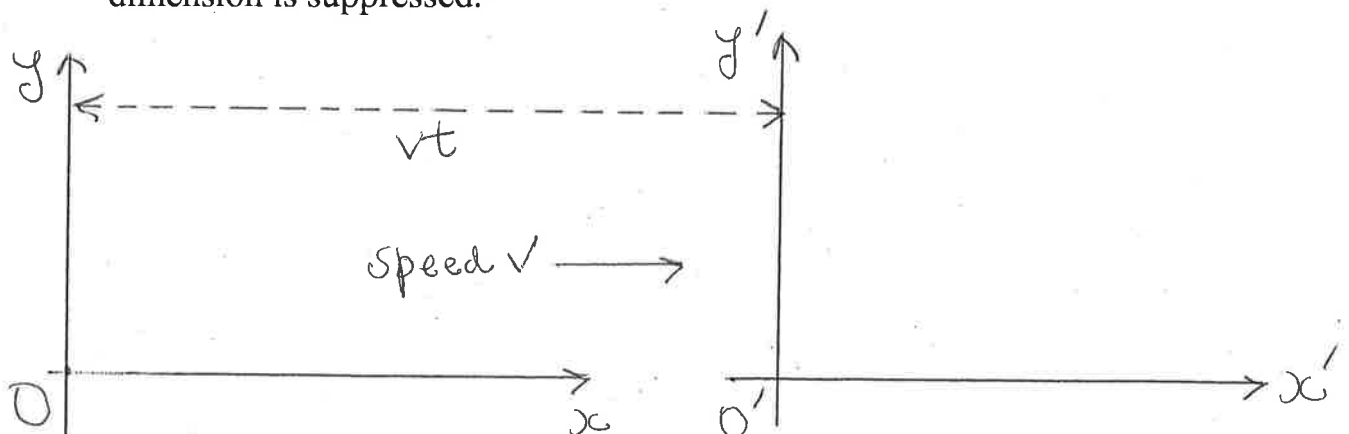
POAMS: LECTURE 8
UNIFORM MOTION: RELATIVISTIC SPEEDS AND
RELATIVISTIC KINEMATICS

I shall now examine some further consequences of Lorentz transformations and time-dilation. It is not my intention to reproduce all the standard results obtainable in Special Relativity in these lectures. I will concentrate on the areas which are relevant to Viv and I's work. In particular, this lecture concerns relativistic speeds and relativistic kinematics, and is based on some of 'Light Speed...' Chapter 4.

1. The special Lorentz transformation

Recall that in the last lecture, I demonstrated how the *special Lorentz transformation*, which relates spatial positions and times, as recorded by different *inertial* observers, may be derived in Normal Realist terms. I derived this transformation directly from the time-dilation formula, without any of the 'electrodynamical' rationale of the standard derivation in Special Relativity, and in particular, without Einstein's second postulate that light travels in *vacuo* at a speed which is constant relative to all differently moving observers.

Consider two observers, O and O' , each situated at the origin of a set of Cartesian axes labelled x, y and z , and x', y' and z' respectively. Suppose that O' moves along the x -axis in the positive direction with *constant* speed v relative to O , such that the x', y' and z' axes remain parallel to the x, y and z axes respectively. Let t be some passage of time as recorded by a clock in O 's observational frame and let t' be the corresponding passage of time as recorded by a clock in O' 's observational frame. Suppose also that the observers O and O' synchronise their clocks when they coincide at $t = 0$, so that when $t = 0$ at $(0, 0, 0)$ relative to O , we have $t' = 0$ at $(0, 0, 0)$ relative to O' . This situation is indicated in the figure below, in which the z dimension is suppressed.



The figure indicates the position of O'' 's frame relative to O 's, after time t , as recorded by O . We call this the *standard situation* between the inertial observers O and O' . Recall that in this special situation between the two inertial observers, if (t, x, y, z) is an *event* as recorded by O and (t', x', y', z') is the same event relative to O' , then

$$x' = \gamma(x - vt), t' = \gamma(t - vx/c^2), y' = y, z' = z, \gamma = (1 - v^2/c^2)^{-1/2} \quad (\text{SLT})$$

This set of equations is the *special Lorentz transformation* for speed v .

This *standard situation* may be generalised but, in practice, given two inertial observers, we may always choose Cartesian axes in each observer's frame so that the standard situation and hence (SLT) holds.

[End of recap]

Recall, from the last lecture, that the *separation*, or *interval* between two events in Minkowski space-time, that is, the right-hand-side of

$$-c^2\tau^2 = -c^2t^2 + x^2 + y^2 + z^2$$

[Equation (5)] where τ is *proper time* and t is *observational time*, provides an *invariant* measure for all inertial observers. This is because both the proper time τ and c in this equation are invariant with respect to all inertial observers. This dictates that this separation must be invariant under any special Lorentz transformation. This is easily checked using equations (SLT) as follows. Relative to the inertial observers O and O' ,

$$\begin{aligned} -c^2t'^2 + x'^2 + y'^2 + z'^2 &= -c^2t'^2 + x'^2 + y^2 + z^2 \\ &= -c^2\gamma^2(t - vx/c^2)^2 + \gamma^2(x - vt)^2 + y^2 + z^2 \\ &= -c^2\gamma^2(t^2 - 2vxt/c^2 + v^2x^2/c^4) + \gamma^2(x^2 - 2xvt + v^2t^2) + y^2 + z^2 \\ &= -c^2\gamma^2t^2(1 - v^2/c^2) + \gamma^2x^2(1 - v^2/c^2) + y^2 + z^2 \\ &= -c^2t^2 + x^2 + y^2 + z^2 \end{aligned}$$

Hence, the separation of two events is invariant under (SLT) as required. In Einstein's Special Relativity, this invariance of the separation is derived first, and then the proper time is defined as a consequence. We believe that ours is the more natural approach.

The applications of (SLT) are many and varied. These are readily available in any text book on Special Relativity. Here, I shall give an application which is of relevance to Viv and I's Normal Realist approach.

2. Relativistic speeds

Consider, once again, the *standard situation* involving inertial observers O and O' , as given in the previous section. Suppose that O and O' observe a moving particle, P , travelling at (not necessarily constant) speed $u(t)$ along the x -axis relative to O . Then P travels along the x' -axis at speed $u'(t')$, say, relative to O' .

According to classical physics, the *Galilean transformation*

$$x' = x - vt, \quad y' = y, \quad z' = z, \quad t' = t$$

[Equation (8), Lecture 7] then gives

$$u'(t') = \frac{dx'}{dt'} = \frac{dx'}{dt} = \frac{dx}{dt} - v = u(t) - v \quad (1)$$

This result supports our everyday experience of relative speeds. For example, if we are travelling down a straight motorway at 80 kilometres per hour and a car overtakes us at 100 kilometres per hour (both relative to the ground), then relative to us, the overtaking car passes us at $100 - 80 = 20$ kilometres per hour. For much higher speeds, however, (SLT) dictates that the classical result, Equation (1), concerning relative speeds, cannot be correct. But of course, we do not experience such high speeds in our everyday lives.

In contrast to classical physics, it follows by the special Lorentz transformation (SLT) that, in this case, since both x' and t' are functions of t ,

$$\frac{dx'}{dt} = \gamma \left(\frac{dx}{dt} - v \right) = \gamma(u(t) - v) \quad (2)$$

and that

$$\frac{dt'}{dt} = \gamma \left(1 - \frac{v}{c^2} \frac{dx}{dt} \right) = \gamma \left(1 - \frac{vu(t)}{c^2} \right) \quad (3)$$

Since $v < c$ and $u(t) < c$ for all t , it follows by the chain rule for differentiation and Equations (2) and (3) that

$$u'(t') = \frac{dx'}{dt'} = \frac{dx'}{dt} \frac{dt}{dt'} = \frac{dx'}{dt} \left(\frac{dt'}{dt} \right)^{-1} = \frac{\gamma(u(t) - v)}{\gamma(1 - vu(t)/c^2)}$$

so that

$$u'(t') = \frac{u(t) - v}{1 - vu(t)/c^2} \quad (4)$$

Equation (4) is the formula for *relativistic subtraction of speeds*. When v and $u(t)$ are very much less than c , this result reduces to the classical result, Equation (1). But when v , or when $u(t)$ becomes, an appreciable fraction of c , relativistic effects become important and we no longer have the simple relationship given in equation (1).

Since observer O moves in the negative x' direction with speed v relative to O' , it follows immediately by Equation (4) that

$$u(t) = \frac{u'(t') + v}{1 + vu'(t')/c^2} \quad (5)$$

This result is consistent with solving Equation (4) for $u(t)$ and is the formula for *relativistic addition of speeds*. It follows by Equation (5) that if $0 \leq u'(t') < c$, then $u(t) < c$:

For any two numbers α and β with $0 \leq \alpha < 1$ and $0 \leq \beta < 1$, we have

$$(1 - \alpha)(1 - \beta) > 0 \Rightarrow 1 + \alpha\beta > \alpha + \beta \Rightarrow \frac{\alpha + \beta}{1 + \alpha\beta} < 1$$

and so, with $u'(t') = \alpha c$ and $v = \beta c$ in Equation (5),

$$u(t) = \frac{u'(t') + v}{1 + vu'(t')/c^2} = \frac{(\alpha + \beta)c}{1 + \alpha\beta} < c$$

Hence, two observational speeds cannot be combined to give a ‘speed’ of greater or equal to c . This tells us that c acts as a universal ‘speed limit’ for all material particles, being the least upper bound of all possible speeds, but **not** a speed itself. Although c is not a speed, note that formally, if both $u'(t')$ and v take the value c in Equation (5), then $u(t) = c$. This is in agreement with our Postulate 2, that c is the same ‘conversion factor’ for converting observational distance to observational time for *all* inertial observers.

This example may easily be generalised to the case in which the inertial observers O and O' observe a material particle moving at velocity $\mathbf{u}(t)$ (in any direction) relative to O and at velocity $\mathbf{u}'(t')$ relative to O' . This case is covered in Example 4.3, pp. 65-66, in ‘Light-speed’

As already mentioned, in addition to the examples presented in these lectures, there are many other applications of the special Lorentz transformation. However, some results, which, traditionally, are a consequence of (SLT) are essentially a direct consequence of the time-dilation formula alone, such as the radial Doppler effect discussed in Lecture 6.

3. Some relativistic kinematics

Let O be any (not necessarily inertial) observer and suppose that P is a moving material object which follows a path C in space relative to O . At any instant of time, t , relative to O , the **position vector**, $\mathbf{r}(t)$, of P is defined by

$$\mathbf{r}(t) = (x(t), y(t), z(t)) \quad (6)$$

where x , y and z are Cartesian coordinates for O ’s frame. This position vector gives the position of P in relation to O at any time and as t varies,

$\mathbf{r}(t)$ varies and traces out the path of P . Then $\mathbf{r} = \mathbf{r}(t)$ is the equation of P 's path C , relative to O .

The (observational) **velocity**, $\mathbf{v}(t)$, of P relative to O gives the rate of change of P 's position, so that

$$\mathbf{v}(t) = \frac{d\mathbf{r}}{dt} = \left(\frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right) \quad (7)$$

The **speed** of P relative to O is then $v(t) = \|\mathbf{v}(t)\|$, the magnitude of P 's velocity. [See *Lecture 7*]. The (observational) **acceleration**, $\mathbf{a}(t)$, of P relative to O is the rate of change of P 's velocity, so that

$$\mathbf{a}(t) = \frac{d\mathbf{v}}{dt} = \frac{d^2\mathbf{r}}{dt^2} \quad (8)$$

These are the definitions that occur in *classical kinematics* but it should be noted that they are not satisfactory in a *relativistic* setting since they are observer dependent. In relativity, we need to replace these classical definitions with analogous definitions that are *independent* of the observer. In particular, in Special Relativity, analogous definitions are based on *events* and *proper time*, rather than points in space and observational time:

Let C denote the path (known as a **world-line**) of a material object in Minkowski space-time. The **position vector**, $\mathbf{R}(\tau)$, of any *event* on C , at any instant of *proper time*, τ , is defined by

$$\mathbf{R}(\tau) = (t(\tau), x(\tau), y(\tau), z(\tau)) \quad (9)$$

Then the **four-velocity**, $\mathbf{U}(\tau)$, and **four-acceleration**, $\mathbf{A}(\tau)$, of P at any instant of proper time are defined respectively by

$$\mathbf{U}(\tau) = \frac{d\mathbf{R}}{d\tau} = \left(\frac{dt}{d\tau}, \frac{dx}{d\tau}, \frac{dy}{d\tau}, \frac{dz}{d\tau} \right) \quad (10)$$

and

$$\mathbf{A}(\tau) = \frac{d\mathbf{U}}{d\tau} = \frac{d^2\mathbf{R}}{d\tau^2} \quad (11)$$

Compare the definitions (9), (10) and (11) with the classical definitions (6), (7) and (8). The four-velocity records the rate of change of events relative

to proper time, as compared to the classical (observational) velocity, which records the rate of change of position relative to Newton's absolute time, and similarly for the four-acceleration. These new definitions are clearly independent of the observer, since they are formulated in terms of the observer-independent proper time parameter, that is, the time as recorded by the travelling object's own clock.

Consider a material object, P , travelling at an (observational) velocity $\mathbf{v}(t)$ relative to an *inertial* observer O . It follows by the time-dilation formula, Equation (6), Lecture 7, that

$$\frac{d\tau}{dt} = (1 - v^2(t)/c^2)^{1/2} \quad (12)$$

where $v(t)$ is the speed of P . In contrast, in standard Special Relativity this follows from the metric for Minkowski space-time and the definition of $\mathbf{v}(t)$. Bearing in mind the value of the Lorentz factor γ in the special Lorentz transformation (SLT), we define $\gamma(t)$ in this context by

$$\gamma(t) = (1 - v^2(t)/c^2)^{-1/2} \quad (13a)$$

It then follows from equation (12) that

$$\frac{dt}{d\tau} = \gamma(t) \quad (13b)$$

Then using the definition of four-velocity, Equation (10), definitions (6) and (7), plus Equation (13b), gives,

$$\mathbf{U}(\tau) = \left(\frac{dt}{d\tau}, \frac{d\mathbf{r}}{d\tau} \right) = \frac{dt}{d\tau} \left(1, \frac{d\mathbf{r}}{dt} \right) = \frac{dt}{d\tau} (1, \mathbf{v}(t)) = \gamma(t) (1, \mathbf{v}(t)) \quad (14)$$

where $\gamma(t)$ is given by Equation (13a). Equation (14) provides the fundamental connection between the relativistic four-velocity, $\mathbf{U}(\tau)$, and the observational velocity, $\mathbf{v}(t)$. It is possible to obtain an expression for the four-acceleration, $\mathbf{A}(\tau)$, in terms of the observational velocity $\mathbf{v}(t)$ and acceleration $\mathbf{a}(t)$, by differentiating Equation (14) with respect to τ and using Equation (13b) again. However, there is no simple relationship between $\mathbf{A}(\tau)$ and $\mathbf{a}(t)$.

In the next lecture, I shall present some relativistic dynamics, and, in particular, we shall study linear momentum, force, mass and energy. It turns out that the fundamental relationship, Equation (14), between the relativistic four-velocity and the observational velocity, is key in the development of relativistic dynamics.

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